GIT-207

# Diarrhea

## Microbiology & Immunology Department

By

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# Objectives:

***By the end of this lecture, the student will be able to:***

- Identify the differences between inflammatory and non-inflammatory diarrhea.
- List the most common infectious causes of inflammatory diarrhea.
- Describe the features, mode of transmission and laboratory diagnosis of infectious causes of inflammatory diarrhea.
- List the most common infectious causes of non-inflammatory diarrhea.
- Explain the features, mode of transmission and laboratory diagnosis of infectious causes of non-inflammatory diarrhea.

# Infective Diarrhea

## Non-inflammatory diarrhea

- There is no mucosal invasion by the organism.
- No PMNs in stool.
- Usually, no fever or mild fever.

## Inflammatory diarrhea (Dysentery)

- There is mucosal invasion by the organism.
- There is PMNs in stool.
- Usually, there is fever.

## Organisms Causing Non-inflammatory Diarrhea

### Viral pathogens

1. Rotaviruses
2. Adenovirus (serotypes 40 & 41)
3. Norwalk virus

### Bacterial pathogens

1. *E. coli* (Enteropathogenic, Enterotoxigenic, Enterohemorrhagic and Enteroaggregative)
2. *Vibrio cholerae*
3. *Vibrio parahaemolyticus*

## Organisms Causing inflammatory Diarrhea (Dysentery)

1. *Campylobacter jejuni*
2. *Salmonella* (subspecies other than typhi)
3. *Shigella*
4. *Yersinia enterocolitica*
5. *Clostridium difficile*
6. Enteroinvasive *E. coli*

# Viral pathogens causing non-inflammatory diarrhea

| | |
|---|---|
| **Rotavirus** | ▪ Non-enveloped, icosahedral **Double-stranded RNA** virus of Reoviridae family.<br>▪ the **most important cause** of gastroenteritis in **infants and young children** (in daycare centers).<br>▪ Transmitted by Fecal-oral route. |
| **Adenovirus (serotypes 40 and 41)** | ▪ Non-enveloped, icosahedral **Double-stranded DNA** virus of Adenoviridae family<br>▪ Causes infection in **young children** (in daycare centers)<br>▪ Transmitted by fecal-oral |
| **Norwalk virus (Norovirus)** | ▪ Non-enveloped, icosahedral **+ss RNA** virus of Caliciviridae family<br>▪ Causes infection in **older kids** (school-aged child) and adults<br>▪ Transmitted by fecal-oral route |

# Bacterial pathogens causing non-inflammatory diarrhea

## *E. coli*

- Gram-negative, facultative anaerobic lactose fermenting bacilli.
- Five types of *E. coli* are known:

1. Enteropthogenic *E. coli* (EPEC)
2. Enteroaggregative *E. coli* (EAEC)
3. Enterotoxigenic *E. coli* (ETEC)
4. Enteroinvasive *E. coli* (EIEC)
5. Enterohemorrhagic *E. coli* (EHEC)

- Enteropathogenic, Enterotoxigenic, Enterohemorrhagic and Enteroaggregative *E. coli* cause non-inflammatory diarrhea.

- Enteroinvasive *E. coli* cause inflammatory diarrhea.
- *E. coli* are present as a part of normal flora in human colon: transmitted by fecal-oral route.

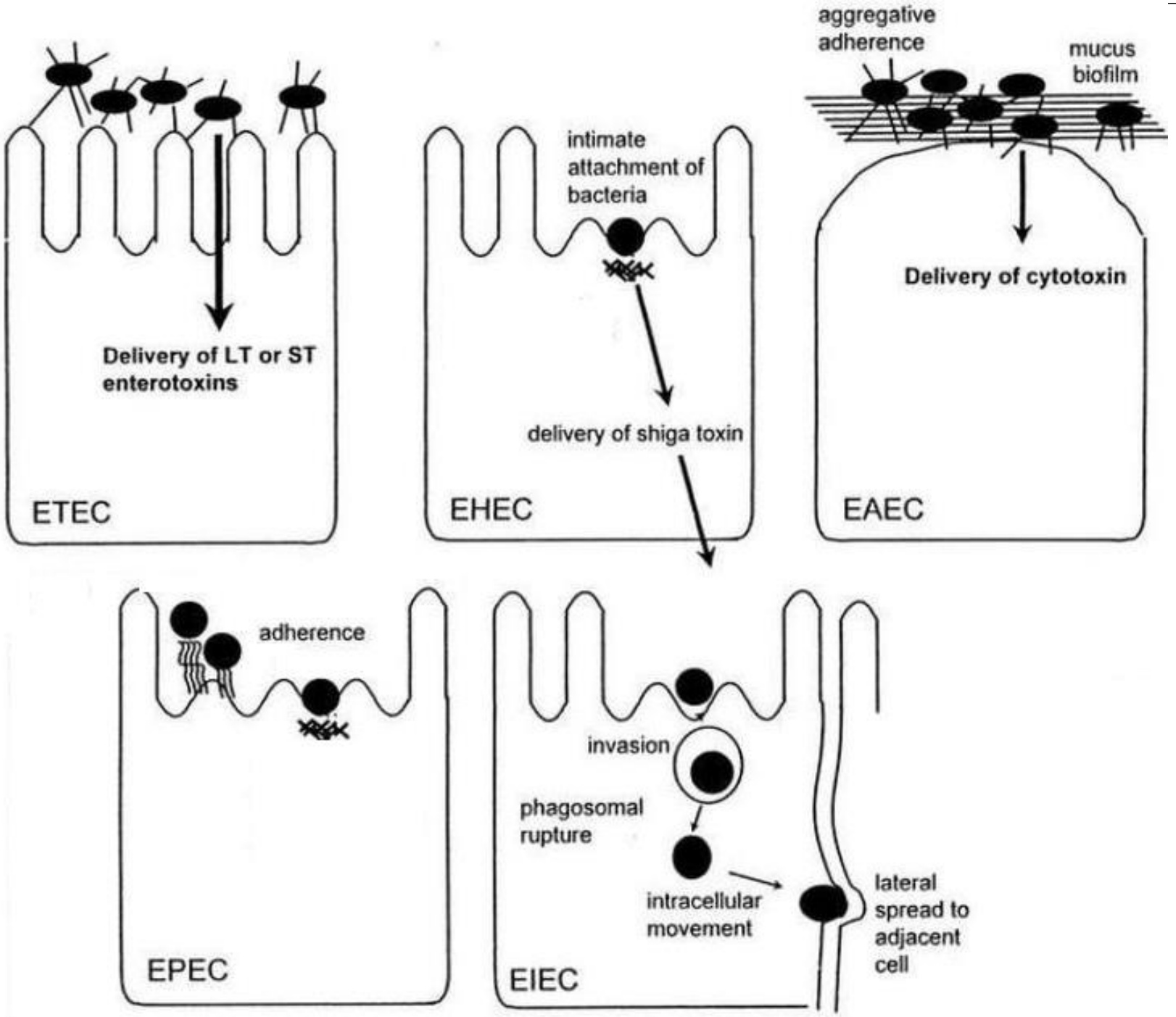
## Enterohemorrhagic *E. coli* (EHEC)

- ***O157:H7 serotype is the most common***
- It is present in bovine colon: transmitted from contaminated food
- It produces verotoxin (shiga-like toxin): causing non-inflammatory bloody diarrhea

## Complications of EHEC

1. Hemorrhagic colitis.
2. Hemolytic uremic syndrome (HUS), because the verotoxin can also bind to the glomerular epithelial cells.

Antibiotics may increase risk of hemolytic- uremic syndrome


aggregative adherence
mucus biofilm
Delivery of LT or ST enterotoxins
ETEC
intimate attachment of bacteria
delivery of shiga toxin
EHEC
Delivery of cytotoxin
EAEC
adherence
EPEC
invasion
phagosomal rupture
intracellular movement
lateral spread to adjacent cell
EIEC

# *Vibrio cholerae*

## Features:

- Gram-negative comma-shaped bacilli with polar flagella → "shooting-star" motility
- Oxidase-positive, catalase-positive and needs alkaline pH for growth.

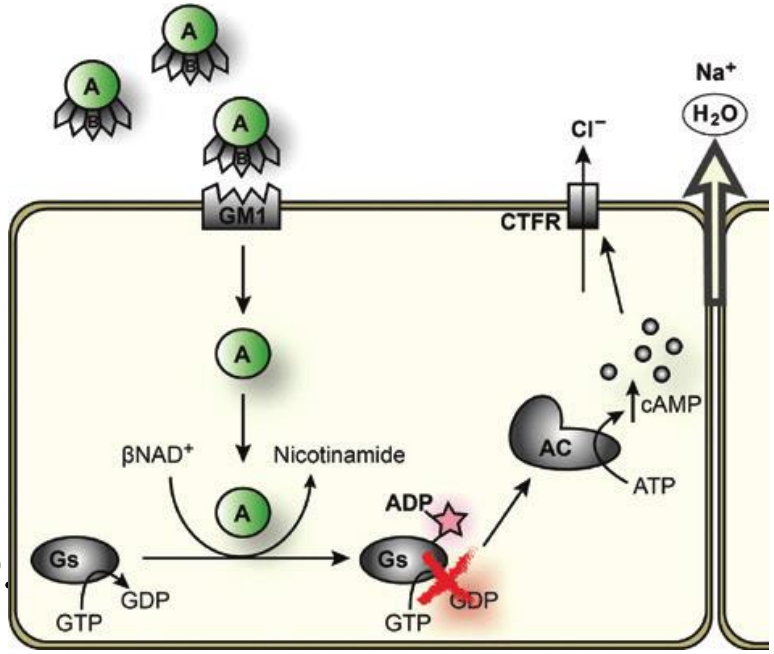
**Reservoir:** Human colon

**Transmission:** contaminated food and water (Fecal-oral)

## Pathogenesis:

It produces enterotoxin that stimulate adenylate cyclase

→ increased cAMP in small intestine → efflux of $Cl^-$ & $H_2O$

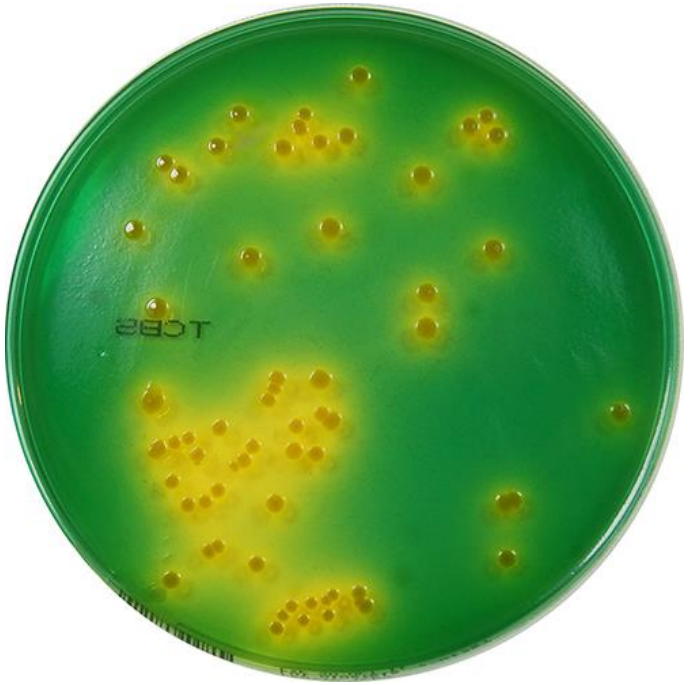


**Disease:** cholera

- Incubation Period: 9–72 hours
- Non-inflammatory profuse watery diarrhea (rice water stools).
- Fever may be present
- Dehydration and hypovolemic shock if not treated.

**Diagnosis:**

- History of travel to endemic area:
- Culture stool on TCBS → yellow colonies & Oxidase positive




**Prevention:** Proper sanitation & Vaccine

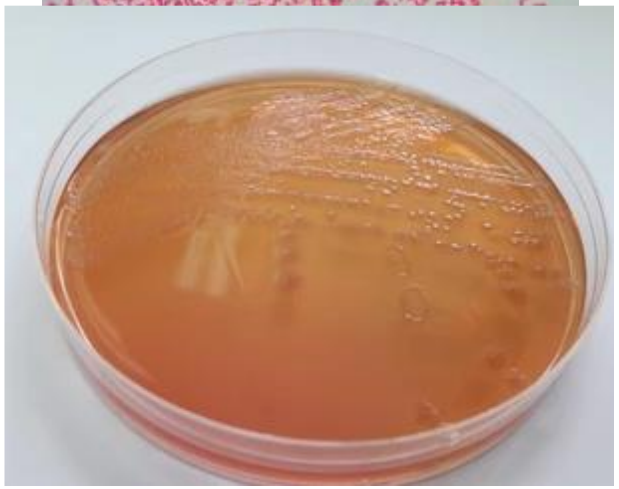
## *Vibrio parahaemolyticus*

- Gram-negative, **Catalase-negative,** oxidase positive comma shaped bacilli with "shooting-star" motility
- Transmitted from raw or undercooked seafood (shellfish)
- Causes self-limited gastroenteritis mimicking cholera

# Common organisms Causing inflammatory Diarrhea (Dysentery)

## *Shigella*

Gram-negative, non-motile, non-lactose fermenting bacilli



**Reservoir:** Human colon only (no animal carriers)

**Transmission:** Fecal-oral spread

**Pathogenesis:**

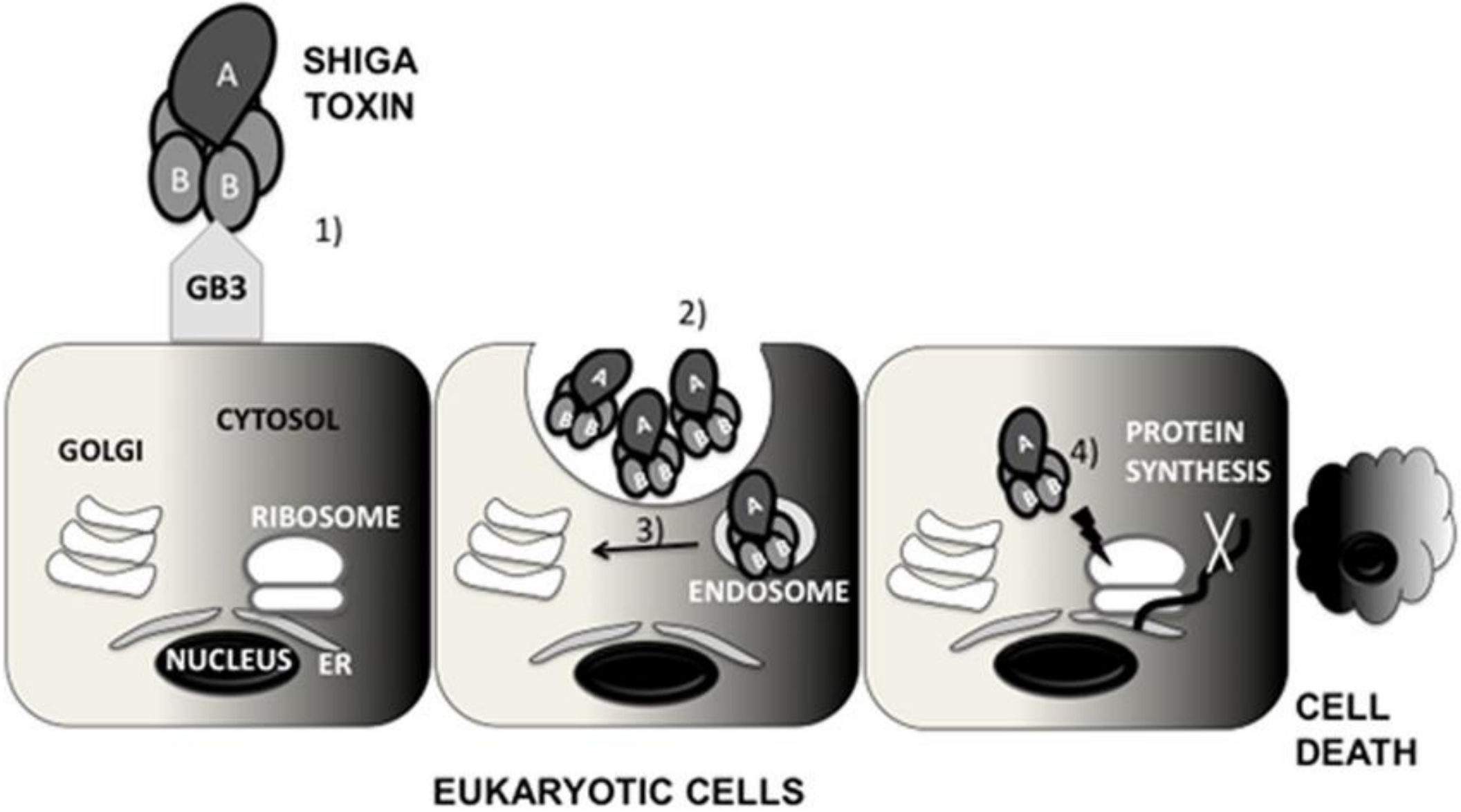
- Shigella colonize small intestine producing enterotoxin and invade the colon mucosa producing shallow mucosal ulcerations and dysentery

- *S. dysenteriae* type 1 produces Shiga toxin:
  - It acts as neurotoxic, cytotoxic and enterotoxic.
  - inhibits protein synthesis in human cells by affecting 60S ribosomal subunit.
  - It can lead to hemolytic uremic syndrome.

**Disease(s):** enterocolitis/shigellosis (most severe form is dysentery)

Watery diarrhea at first → lower abdominal cramps, tenesmus and abundant pus and blood in the stools (dysentery).

# Mechanism of action of Shiga Toxin



# *Clostridium difficile*

Gram-positive, spore-forming, anaerobic bacilli.

**Reservoir:** Human colon/gastrointestinal tract

Infection occurs from patient's own flora (**endogenous source**) in hospitalized patients on antibiotics.

**Pathogenesis:**

- Antibiotics change the composition and function of the intestinal flora → over-growth of *Cl. dificile*
- *Cl. dificile* secretes enterotoxin that damage the colonic mucosa leading to pseudo-membranous colitis

## Clinically

- Mild to severe inflammatory diarrhea, abdominal cramps, fever.
- Blood, mucus, and pus in stools

## Diagnosis:

- Stool exam for toxin production.
- Culture is not diagnostic because organism is part of normal flora.

## Prevention:

- Use caution in overprescribing broad-spectrum antibiotics (consider limited spectrum drugs first).

ANY
QUESTIONS?